

Supplemental Material: Observing the quantum Mpemba effect in quantum simulations

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A. Experimental apparatus

In this work, we used a trapped-ion quantum simulator capable of storing and manipulating long ion strings for studying the QMPE. The experimental setup consists of a string of $N = 12$ $^{40}\text{Ca}^+$ ions that is made to emulate a spin-1/2 chain with long-range interactions. Each qubit is labeled by basis vectors $s = (|\downarrow\rangle, |\uparrow\rangle) \equiv (|1\rangle, |0\rangle)$, which are encoded onto $|S_{1/2}, m = +1/2\rangle$ and $|D_{5/2}, m = +5/2\rangle$ electronic levels of a calcium ion. The power-law XY interaction expressed in Eq. 2 (see main manuscript) is engineered with the help of a global laser beam that interacts with ions' electronic and motional degrees of freedom from the transverse direction of the ion chain [1, 2]. Furthermore, the disordered Hamiltonian displayed in Eq. 3 (see main manuscript) is generated by shining a set of site-selective light shift beams along with the global entangling beam [3]. In the experimental setup, we have noticed that the time-evolved states experience dephasing while undergoing the time evolution. The dephasing rates have been measured for all cases presented in this letter. The coherence times,

which are related to dephasing rates via $\Gamma = 1/(2T_{\text{coh}})$, are $T_{\text{coh}} \sim 16$ ms for the XY interaction case, $T_{\text{coh}} \sim 45$ ms for the pure dephasing cases, $T_{\text{coh}} \sim 27(2)$ ms for the strong disorders and $T_{\text{coh}} \sim 27(4)$ ms for the weak disorder cases.

B. AC-Stark shift compensation during the spin-spin interaction

The trapped-ion platforms employ an Ising interaction when a bichromatic laser beam off-resonantly excites the motional red and blue sidebands of an ion chain [2]. In the case of symmetric detuning Δ of the bichromatic beam from the motional sidebands, an AC-Stark shift is present due to the coupling of the laser to the ion's spectator levels. For example, in our setup, when the interaction is generated on the quadrupole transition of a calcium ion chain with a 729-nm laser beam, the off-resonant coupling to the dipole-allowed transitions gives an AC-Stark shift. We cancel this shift by adding a third frequency component to the laser beam, such that the light off-resonantly couples to the quadrupole transition and induces an AC-Stark shift that is of opposite sign but of equal strength compared to the dipole transitions.

The Ising interaction is then transformed into an XY interaction via asymmetrically driving the sidebands by $\Delta \pm \delta_c$. In the limit of δ_c being smaller than Δ and much larger than the spin-spin coupling strength J_0 , the interaction Hamiltonian is written as [4],

$$H = \sum_{i>j} \frac{J_0}{2|i-j|^\alpha} (\sigma_i^x \sigma_j^x + \sigma_j^y \sigma_i^y) + \frac{\delta_c}{2\Delta} \sum_i B_i \sigma_i^z, \quad (1)$$

where an additional AC-Stark shift proportional to δ_c is induced. This term is often irrelevant when the input states are initialized in the eigenbasis of the σ_z operator. However, it is relevant when the input state is not the eigenstate of the σ_z operator, which is currently the case for $\theta \neq 0$. In the above equation, B_i depends upon the motional occupation number $\langle N_\ell \rangle$, site-dependent Rabi frequency Ω_i and mode dependent Lamb-Dicke

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parameter $\eta_{i,\ell}$; and it is expressed as

$$B_i = \Omega_i^2 \Delta \sum_{\ell} \frac{\eta_{i,\ell}^2}{\Delta_{\ell}^2} (\langle N_{\ell} \rangle + 1/2). \quad (2)$$

Here, Δ_{ℓ} is the laser detuning from the ℓ^{th} motional sideband. The shift can be different for each ion due to its dependence on the Lamb-Dicke parameter which is different for each ion. However, the major contribution comes from the center of mass mode, which is closest to the laser drive frequency, and it is eliminated by appropriately adjusting the intensity of the third frequency component of the laser beam.

C. Experimental noise

The dominant decoherence effect in our experiment is that of dephasing noise, an unwanted noise that randomly rotates the spin vector along the z axis. In our system, the dephasing is a combination of laser phase noise/frequency noise and magnetic-field noise. We measure the overall effect of noise by performing Ramsey measurements for entangling interaction, disorder, and pure dephasing cases. For the XY quenches, the measured coherence time is $T_{\text{coh}} \sim 16$ ms. The coherence time here is measured such that we could account for laser phase noise, magnetic field noise, and fluctuating AC-Stark shift due to thermal occupation of the motional modes. Specifically, the measured coherence time without turning on the quench beam is $T_{\text{coh}} \sim 45$ ms.

A shorter coherence time in the case of entangling interactions can be explained by considering the thermal occupation of the motional modes. While writing this manuscript we became aware of this issue and carried out some detailed experimental and numerical studies beyond the scope of the present manuscript and verified that the drop in the coherence time is indeed due to this effect. Its impact can be understood from the second term of Eq. (1). The AC-Stark shift in this case linearly follows the phonon number, thus imperfect ground-state cooling or motional heating increases the dephasing rates in the system. To improve this coherence, we are currently working on implementing better ground-state cooling methods in our system and designing a new ion trap to reduce the motional heating of the ion string.

In the disorder studies, we use a spin-echo sequence to mitigate slow drifts in the experimental parameters. To do so, an echo pulse, i.e. a π rotation about the x axis in the middle of the time evolution is applied. The coherence time measured is $T_{\text{coh}} \sim 27(2)$ ms for strong disorders and $T_{\text{coh}} \sim 27(4)$ ms for weak disorders. In the case of pure dephasing, where the prepared state is left to evolve under dephasing noise without interactions, we observe that slow detrimental noise components affect the quality of randomized measurements. To overcome

this, we again use a spin-echo sequence. The measured coherence time is $T_{\text{coh}} \sim 45(4)$ ms.

In the numerical simulations shown in the results, we have considered a global dephasing mechanism with the experimentally measured rates. The corresponding jump operator is $\sqrt{\Gamma} \sum_i \sigma_i^z$, where $\Gamma = 1/(2T_{\text{coh}})$. The spontaneous decay rate and depolarization noise rate reported elsewhere [5] are negligible compared to these rates and hence they are not the leading error sources for the current studies. To counteract slow drifts of the experimental parameters, the system is calibrated every few minutes for changes in clock transition, laser pulse area fluctuation, and shift in the ion position.

The impact of decoherence in observing the QMPE is further studied via numerical simulations. The results are displayed in Fig. 1 while considering various values of global dephasing rates Γ . We notice that features like the crossing of EA curves and observation of the QMPE remain unaffected. However, dephasing decreases EA for all tilt angles. The revival of EA is noticeably damped down by dephasing (see the peak at around $t = 7.5$ ms). Nonetheless, we can conclude that for qualitative purposes the QMPE is robust under global dephasing.

D. Implementation of disorder in the experiment

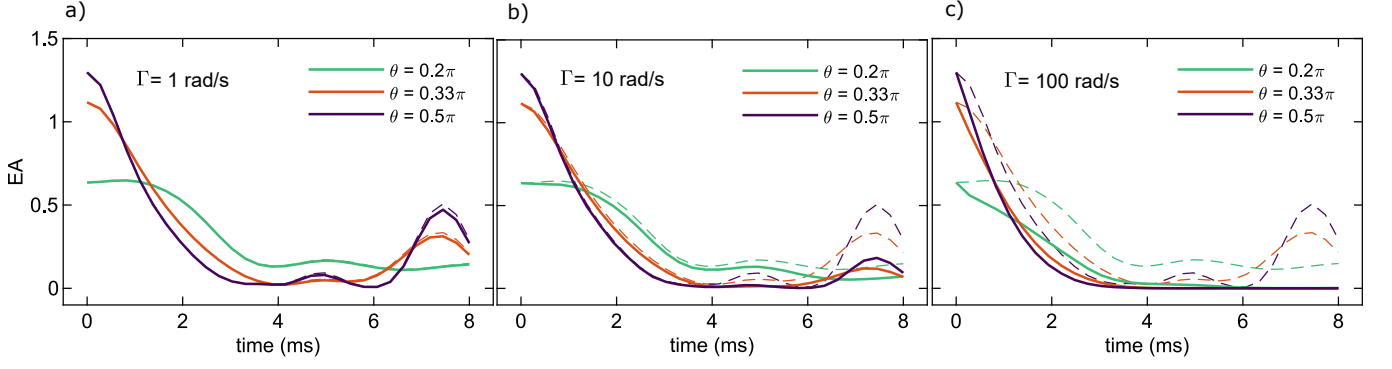
For the disorder studies presented in the main manuscript, the XY interaction is combined with site-dependent transverse field terms. This disorder Hamiltonian reads as

$$H = \sum_{i>j} \frac{J_0}{2|i-j|^{\alpha}} (\sigma_i^x \sigma_j^x + \sigma_j^y \sigma_i^y) + \sum_i h_i \sigma_i^z, \quad (3)$$

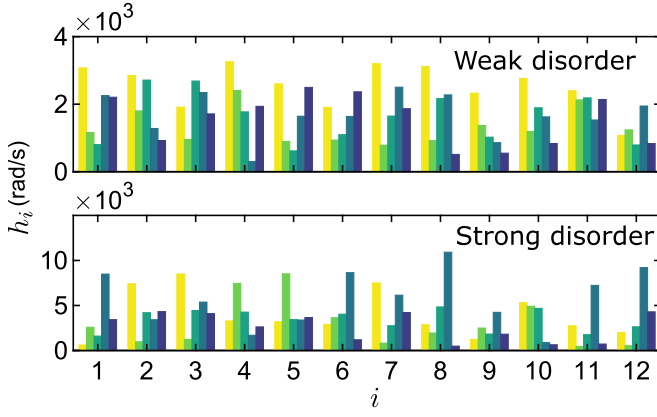
where h_i indicates transverse disorder strength. In our experimental system, these disorders terms are realized by shining multiple laser beams generated by an acousto-optic deflector (AOD) on individual ions. The AOD deflects multiple laser beams designated to individual ions with the desired light intensity such that site-dependent AC-Stark shifts are induced. We sample 5 sets of weak and strong disorders randomly chosen from a uniform distribution function and apply them to the ion string. In Fig. 2, we show the measured disorders for both the strong and weak disorder cases.

E. Measurements

The measurement protocol is shown in Fig. 1a of the main manuscript. We begin by preparing a ferromagnetic product state $\rho_{\text{ferro}} = (|\downarrow\rangle\langle\downarrow|)^{\otimes N} \equiv (|1\rangle\langle 1|)^{\otimes N}$. This state is $U(1)$ symmetric under rotations about the z axis. Next, we create initial states that explicitly break the $U(1)$ symmetry by rotating the state ρ_{ferro} about the $-y$ axis using the operator $R_{-y}(\theta) = \exp(i\theta S_y/2)$ with $S_y = \sum_i \sigma_i^y$. This allows us to create the symmetry breaking



SM Figure 1: We numerically examine the effect of global dephasing on EA and on the observation of the QMPE. Dashed lines correspond to unitary dynamics and solid lines correspond to dephasing dynamics. With increasing dephasing rates; a) $\Gamma = 1$ rad/s, b) $\Gamma = 10$ rad/s, and c) $\Gamma = 100$ rad/s, a reduction in EA is observed for all three tilt angles. Conversely, the QMPE feature i.e. crossing of EA curves for three tilt angles is present even for the highest dephasing rate.



SM Figure 2: Disorder strength h_i versus the ion index for the weak and strong disorder realizations employed in the experiments. Each color corresponds to one of the 5 sets of different realizations of disorders considered.

initial states $\rho(\theta)_{\text{init}} = R_{-y}(\theta)\rho_{\text{ferro}}R_y(\theta)$, where the tilt angle θ controls the extent of broken symmetry. This state is quenched with Hamiltonians of interest [Eqs. 2, 3 of the main manuscript] for a duration t , leading to the state $\rho(\theta, t)$ that is finally measured in random bases (detailed below). The measurements are performed for each tilt angle θ and time t independently, thus, hereon we drop the parenthesis to be concise.

The randomized measurement (RM) framework [6, 7] allows us to estimate desired quantities in a state-agnostic manner with a reduced measurement budget compared to the quantum state tomography [8]. This toolbox has allowed probing a large range of interesting quantum properties such as entanglement entropies [9–12], negativities [13–15], bound entanglement [16, 17], quantum Fisher information [18, 19] and many-body quantum chaos [20]. The RM protocol requires applying unitaries $U = \bigotimes_{i=1}^N U_i$ to the time-evolved states, where each single qubit random rotation U_i is sampled from

at least a 2-design. Then, the rotated state $U\rho U^\dagger$ is projected in the z basis $|\mathbf{s}\rangle = |s_1, \dots, s_N\rangle$ with $s_i \in (0, 1)$. Note that, in the experiment, we construct local random unitaries U_i by combining global x and y rotations with single qubit local z rotations whose angles are sampled randomly to obtain a set of unitaries from the circularly uniform ensemble (CUE) [21]. For the x and y global rotations, we use an elliptically shaped laser beam coupling to all ions simultaneously and driving the resonant optical transition between spin states. A tightly focused laser beam is used to perform local z rotations, by inducing an AC-Stark shift of variable strength on individual ions. Quantum projective measurements in the z basis are obtained via site-resolved fluorescence detection.

The protocol described above is repeated for N_U different random unitaries and recording N_M projective measurements for each random unitary. The experimental data thus obtained comprises of random unitaries $U^{(r)}$ and the respective recorded bitstring measurements $\mathbf{s}^{(r,m)} = (s_1^{(r,m)}, \dots, s_N^{(r,m)})$ with $r = 1, \dots, N_U$ and $m = 1, \dots, N_M$ that are stored in a classical device for post-processing. From the collected RM dataset, we can construct operators known as *classical shadows* of the prepared state ρ in the post-processing stage. This operator restricted to a given subsystem A can be defined as [6, 22]

$$\hat{\rho}_A^{(r,m)} = \bigotimes_{i \in A} 3 U_i^{(r)\dagger} |s_i^{(r,m)}\rangle \langle s_i^{(r,m)}| U_i^{(r)} - \mathbb{I}_2. \quad (4)$$

These classical shadows form an unbiased estimator of the underlying density matrix of interest, that is, the average over the applied unitaries and the measurement outcomes gives $\mathbb{E}[\hat{\rho}_A^{(r,m)}] = \rho_A$.

F. Estimation of the EA

The shadows defined in Eq. (4) can be used to extract arbitrary multi-copy functionals of the form $f_n = \text{Tr}(O_A^{(n)} \rho_A^{\otimes n})$ defined on the subsystem A using the U-statistics estimator [23]:

$$\hat{f}_n = \frac{1}{n!} \binom{N_U}{n}^{-1} \sum_{r_1 \neq \dots \neq r_n} \text{Tr} \left(O_A^{(n)} \bigotimes_{j=1}^n \hat{\rho}_A^{(r_j)} \right), \quad (5)$$

which is an unbiased estimator of f_n , i.e. $\mathbb{E}[\hat{f}_n] = f_n$. We also define here the classical shadow $\hat{\rho}_A^{(r)} = \mathbb{E}_{N_M}[\hat{\rho}_A^{(r,m)}]$, constructed by averaging over N_M measured bitstrings for an applied unitary $U^{(r)}$. Additionally, the classical shadow formalism allows one to estimate interesting symmetry-resolved quantum properties of the underlying quantum state as shown in the Refs. [15, 24, 25]. In this work, in order to obtain an estimator $\widehat{\text{EA}}$ of Eq. 1 (main manuscript) from the RM dataset, we construct unbiased estimators of each of the individual trace moment term in Eq. 1 (main manuscript) following Refs. [6, 24]. This is written as,

$$\widehat{\Delta S}_A = \log \frac{\sum_{r_1 \neq r_2} \text{Tr}(\hat{\rho}_A^{(r_1)} \hat{\rho}_A^{(r_2)})}{\sum_{r_1 \neq r_2} \text{Tr}(\hat{\rho}_{A,Q}^{(r_1)} \hat{\rho}_{A,Q}^{(r_2)})}, \quad (6)$$

where $\hat{\rho}_{A,Q}^{(r)}$ is the shadow for the symmetrized counterpart of $\hat{\rho}_A^{(r)}$ given by $\hat{\rho}_{A,Q}^{(r)} = \sum_{q \in \mathbb{Z}} \Pi_q \hat{\rho}_A^{(r)} \Pi_q$ that satisfies $\mathbb{E}[\hat{\rho}_{A,Q}^{(r)}] = \rho_{A,Q}$. Rigorous performance analysis of statistical errors from a finite number of experimental runs $N_U \times N_M$ in the estimation of such trace moments have been studied in various prior works [14, 15, 18, 22].

G. Frobenius distance

For generic large quantum systems evolving unitarily, the reduced density matrix describing the subsystem A is expected to relax, at long times, to a Gibbs distribution [26–28]. The eigenstate thermalization hypothesis [26, 27] predicts that such a thermal state is equivalent to the reduced density matrix of the diagonal ensemble [28, 29]. The diagonal ensemble is obtained by projecting the initial density matrix onto the eigenbasis $\{|E_k\rangle\}$ of the quenching Hamiltonian, that is

$$\rho^{\text{DE}}(\theta) = \sum_k \langle E_k | \rho(\theta, t=0) | E_k \rangle | E_k \rangle \langle E_k |, \quad (7)$$

where $\rho(\theta, t=0)$ is the initial state. The partial trace of this state $\rho_A^{\text{DE}}(\theta) = \text{Tr}_{\bar{A}}(\rho^{\text{DE}}(\theta))$ gives the average behavior of local operators at long times. For each tilt angle θ , we are interested in the Frobenius distance between the post-quench state of subsystem A at a given

time and the corresponding diagonal ensemble i.e., the distance $\text{FD}[\rho_A(\theta, t), \rho_A^{\text{DE}}(\theta)]$, defined as [30],

$$\text{FD}(\rho_A, \rho_A^{\text{DE}}) = \sqrt{1 - \min \left(1, \frac{2 \text{Tr}[\rho_A \rho_A^{\text{DE}}]}{\text{Tr}[\rho_A^2 + (\rho_A^{\text{DE}})^2]} \right)}, \quad (8)$$

where we have dropped the angle θ and time t for brevity. Here, we have put a bound on the second term to bypass the unphysical Fidelities. In the limit of infinitely many measurements, the second term should always be upper bound by 1. In the main text, we utilized this distance to analyze the thermalization of the reduced density matrix after the XY Hamiltonian quench (see Fig. 4).

The rich RM dataset obtained using the protocol described above equally allows us to extract the Frobenius distance. Namely, we can estimate unbiased estimators of all the individual terms in the expression of the Frobenius distance (8) from the *same* dataset collected during the experimental execution. Thus the explicit expression of the estimator $\widehat{\text{FD}}$ of the Frobenius distance can be written as

$$\widehat{\text{FD}} = \left(1 - \frac{\frac{2}{N_U} \sum_{r_1} \text{Tr}[\hat{\rho}_A^{(r_1)} \rho_A^{\text{DE}}]}{\beta^{-1} \sum_{r_1 \neq r_2} \text{Tr}[\hat{\rho}_A^{(r_1)} \hat{\rho}_A^{(r_2)}] + \text{Tr}[(\rho_A^{\text{DE}})^2]} \right)^{1/2} \quad (9)$$

with $\beta = N_U(N_U - 1)$.

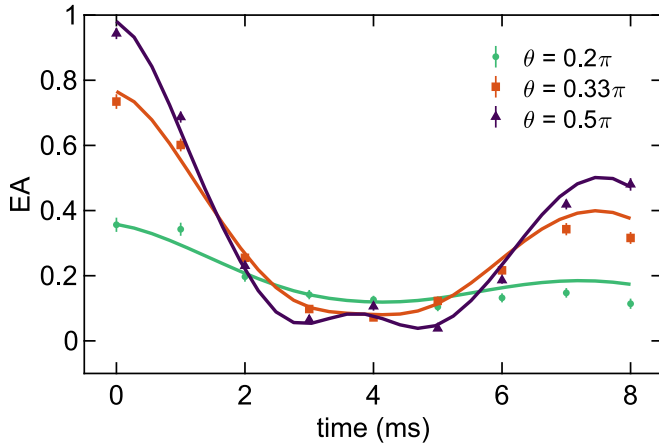
H. Estimation of error bars

For randomized measurements, experiments are carried out for N_U random unitaries and N_M number of projective measurements for each random unitary. In the main text, the time evolution with the XY interaction (Fig. 1b) is done with $N_U = 500$ and $N_M = 30$. For the inset of Fig. 1b, we sample $N_U = 200$ and $N_M = 100$. The experiments in the presence of disorder (results in Fig. 2) and pure dephasing (results in Fig. 3) are done with $N_U = 300$ and $N_M = 30$. We used a total of 5 disorder sets for each θ in the disorder studies.

In the main text, for the disorder case, the error bars are estimated from the standard deviation of the mean over disorder realization. Elsewhere, the error bars are estimated via the Jackknife method applied on the random unitaries that are used to estimate the averaged EA [6, 31]. Wherever the subsystem average data are presented, we employ the Jackknife resampling method by dropping one random unitary at a time and subsequently estimating the variance of the subsystem averaged EA. The subsequent error bars of the mean are presented in the figures.

I. Finite size effects: 4-qubit experiment

To explore the finite-size effects within the experiment's coherence time, we measure the EA



SM Figure 3: The EA after a quench with the XY Hamiltonian on a system of $N = 4$ qubits. The averaged EA over all possible subsystems of size $N_A = 2$ is plotted for three tilt angles. Symbols are the experimental data and curves correspond to numerical simulations. The subsystem restores the symmetry and the QMPE is observed. However, a revival of the EA is seen due to the finite-size effects.

for a 4-qubit system and present the results of the EA for $N_A = 2$ subsystems. Data are plotted in Fig. 3. Here, symbols correspond to experimental data, and solid lines correspond to numerical simulations in the presence of global dephasing. It is important to note that QMPE is present even for the smallest system that we probed in the experiment. However, a revival of the EA is observed due to finite-size effects. In fact, in this case, thermalization is hindered and thus the subsystem oscillates between the asymmetric and symmetric states.

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